

Reflective Journal

Mastering the Machine Learning Workflow

COURSE	ITAI 1371 — Classic Machine Learning
ASSIGNMENT	Module 03 Lab Exercise, Reflective Journal (Cell 25)
SUBJECT	Wine Classification — Supervised Learning Models
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1. Supervised, Unsupervised, and Reinforcement Learning

These are three different ways a computer can learn something, and the difference comes down to what kind of information it is given to start with.

Supervised learning is when you give the computer a set of examples where you already know the correct answer, so it can check its own work as it learns. It is the equivalent of studying with a practice test that has the answer key attached.

Unsupervised learning is when you hand the computer a pile of information with no labels at all, and it has to find its own patterns and group things together on its own. It is similar to sorting a mixed bag of candy by color or texture without anyone telling you the brand names.

Reinforcement learning is when a computer learns through trial and error, trying different moves, earning points for good outcomes and losing points for bad ones, the same way a player improves at a video game just by playing it repeatedly.

2. Best Performing Model

Top Performer	Final Accuracy
Logistic Regression	0.917

3. Why It Performed Better

Logistic Regression draws one clean, straight boundary to separate the classes. Because our data was already cleaned and standardized ahead of training, that simple, stable approach held up well against new data.

The Decision Tree scored lower despite showing no overfitting itself. Its own learning curve gap was 0.038, the same healthy range as Logistic Regression's 0.035, and it was built with a max depth of 3, which limits how complex its rules are allowed to get. So the gap in accuracy is not an overfitting problem. It comes down to shape. A depth-limited tree can only draw a small number of straight, boxy

splits, and that boundary is simply a worse fit for how these three wine classes actually separate than the single smooth line Logistic Regression draws.

4. What I Would Try Next

- **K-Fold Cross-Validation:** run the evaluation multiple times on different slices of the data, rather than a single split, to confirm the 0.917 score is consistent and not just a lucky result.
 - **Hyperparameter Tuning:** adjust the model's internal settings to see whether accuracy can be pushed even higher.
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5. Real-World Application

Industry	Agriculture
ML Problem	Predicting crop health early enough to act on it
Type of ML	Supervised learning (classification)
Data Needed	Soil moisture and temperature readings

A Logistic Regression model could analyze soil moisture and temperature readings to predict crop health early. Farmers could then apply water or nutrients only where they are actually needed, instead of treating an entire field the same way.

6. Most Important Concept Learned

Generalization. A model is not actually good simply because it scores well on the data it already studied. It is only good if it can perform on new data it has never seen. A perfect score on training data means nothing if the model fails once it meets the real world.

7. Most Challenging Part

Avoiding data leakage. The data had to be split into training and testing groups before any other step. Values such as the average were calculated using only the training portion, then applied to the test set using those same numbers. Scaling the full dataset before splitting would let the model see information from the test set in advance, making the results look better than they actually are.

8. Question for Further Exploration

How might we ensure our models stay fair and ethical if the labeled data used to train them already contains hidden human bias?